# UNIT – V

## Creep Behaviour & Testing

Subject : Mechanical Behaviour of Materials (23A03504c)

> Syllabus: Creep curve and its stages, structural changes during creep, creep mechanisms, metallurgical factors affecting creep, high-temperature alloys, stress-rupture testing, creep testing machines.

## 1. Introduction – Creep

**Creep** = time-dependent, permanent deformation of a material under a constant stress (or load) that is usually below the yield strength, occurring at high temperature – generally $T > 0.4\ T_m$ (homologous temperature $T/T_m$, $T$ in kelvin).

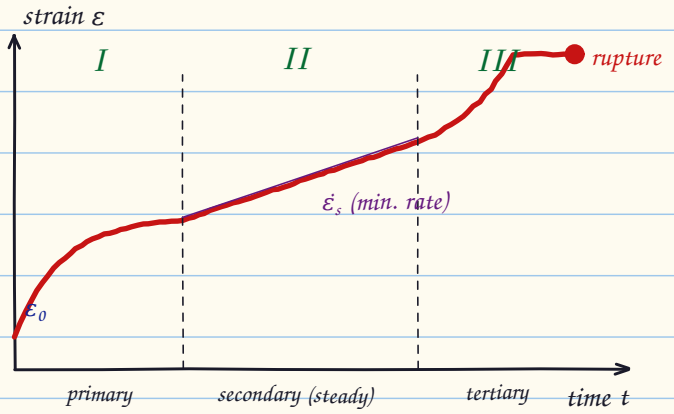
- Examples : turbine blades and discs, boiler/steam pipes and superheater tubes, nuclear fuel cladding, furnace parts, jet engines, rocket nozzles, heat-exchangers, solder joints, lead pipes and roofs at room temperature (Pb $T_m$ = 600 K ⇒ RT = 0.49 $T_m$), tungsten lamp filaments (sagging), glacier flow.
- Critical temperature for creep (0.4 $T_m$) : Fe 451 °C ; Ni 418 °C ; Al ≈ 100 °C (0.4×933 = 373 K) ; Cu 270 °C ; Ti 530 °C ; W 1200 °C ; Pb, Sn ≈ room T ; polymers near RT. Tungsten at 1000 °C (0.35 $T_m$) hardly creeps, while Al at 150 °C does.
- Failure modes in service : excessive deformation (loss of clearance – blade tip rubbing), creep rupture (fracture after long time), stress relaxation of bolts/springs (creep under constant strain), creep-fatigue interaction. Design life $10^5$ h (≈ 11.4 years) for power plant.

| Metal | $T_m$ (K) | 0.4 $T_m$ (K) | 0.4 $T_m$ (°C) |
|---|---|---|---|
| Pb | 600 | 240 | –33 |
| Al | 933 | 373 | 100 |
| Cu | 1356 | 542 | 269 |
| Fe | 1811 | 724 | 451 |
| Ni | 1728 | 691 | 418 |

| Metal | $T_m$ (K) | 0.4 $T_m$ (K) | 0.4 $T_m$ (°C) |
|---|---|---|---|
| Ti | 1941 | 776 | 503 |
| W | 3683 | 1473 | 1200 |

## 2. The Creep Curve

Creep curve (constant load, T)




$\varepsilon = \varepsilon_0 + \varepsilon_t(1 - e^{-rt}) + \dot{\varepsilon}_s\, t$ (Garofalo)

I : decreasing rate (hardening)
II : hardening = recovery
III : necking, voids, rupture

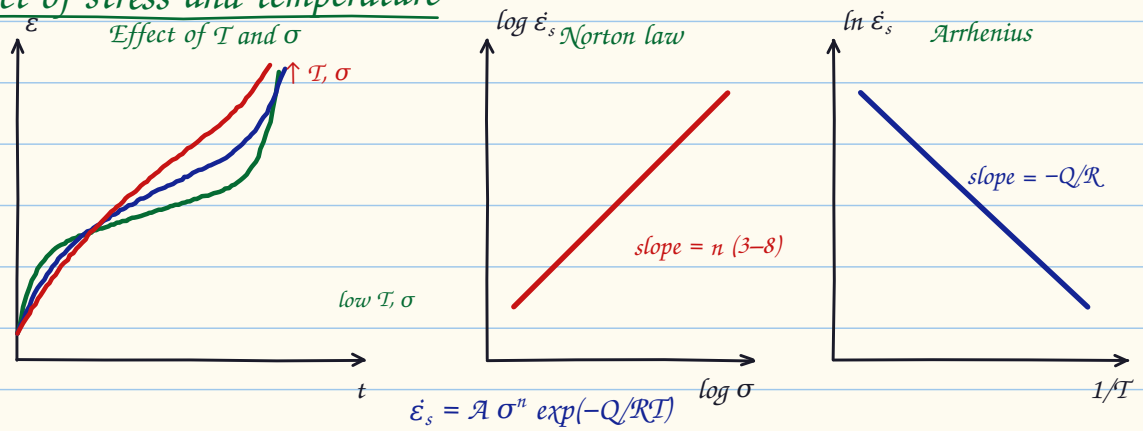
Strain ε is plotted against time t at constant temperature and constant load (stress). Three stages after the instantaneous strain :

- **Instantaneous (initial) strain $\varepsilon_0$** : elastic (+ some plastic) strain on loading.
- **Stage I – primary (transient) creep** : creep rate $\dot{\varepsilon} = d\varepsilon/dt$ decreases with time ; strain hardening dominates – dislocations multiply and tangle, the material becomes more resistant.
- **Stage II – secondary (steady-state, viscous) creep** : approximately constant minimum creep rate $\dot{\varepsilon}_s$ ; a dynamic balance of strain hardening and recovery (softening) – dislocations climb, annihilate, form subgrains. Longest stage, used in design (creep rate × life).
- **Stage III – tertiary creep** : rate accelerates leading to rupture ; caused by necking (area reduction at constant load increases true stress), internal cavities / grain-boundary cracks, microstructural degradation (coarsening of precipitates, over-ageing, recrystallisation) and oxidation.

$\varepsilon = \varepsilon_0 + \varepsilon_t(1 - e^{-rt}) + \dot{\varepsilon}_s\, t$ (Garofalo ; transient + steady) ; low-T logarithmic creep $\varepsilon = \alpha\, \ln(1 + \beta t)$

- **Andrade** : $\varepsilon = \varepsilon_0(1 + \beta\, t^{1/3})\, e^{Kt}$. Constant-stress tests (cam compensation) show a longer secondary stage and no tertiary stage from necking (only structural) ; **constant-load** tests are standard in industry.

## Effect of stress and temperature




$\dot{\varepsilon}_s = A\,\sigma^n \exp(-Q/RT)$

- Higher $T$ or $\sigma$ ⇒ larger $\varepsilon_0$ and $\dot{\varepsilon}_s$, shorter primary stage, earlier tertiary stage, shorter rupture life $t_r$ ; very low $T/\sigma$ ⇒ only primary creep (strain decays, creep stops).
- **Steady-state creep rate** :

$$\dot{\varepsilon}_s = A\,\sigma^n \exp(-Q_c/RT)$$ (Norton power law × Arrhenius ; Mukherjee–Bird–Dorn)

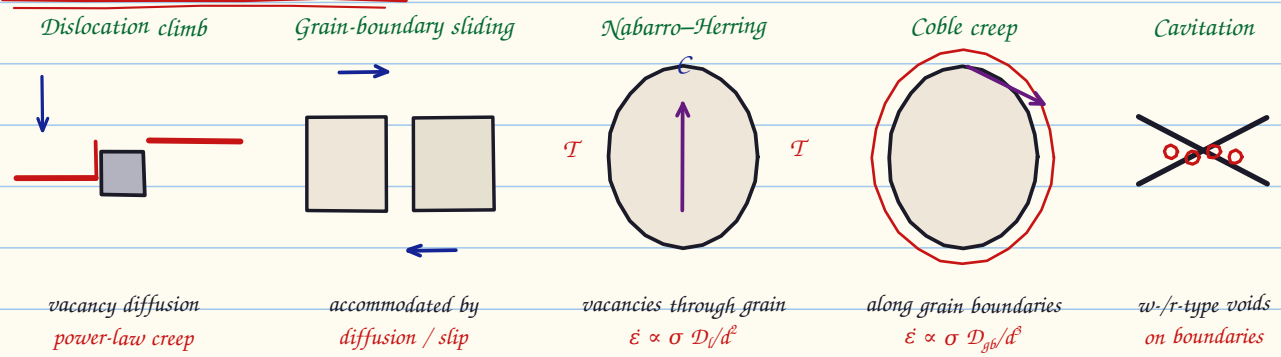
$A$ = constant (structure) ; $n$ = stress exponent (3–8 for dislocation (climb) creep ; 1 for diffusional creep) ; $Q_c$ = activation energy for creep ≈ $Q_{SD}$ (self-diffusion : Al 142, Cu 200, Fe 280, Ni 285 kJ/mol) ; $R$ = 8.314 J/mol·K.

- **Monkman–Grant relation** : $\dot{\varepsilon}_s \cdot t_r$ = constant (≈ 0.01–0.1) ⇒ rupture life inversely proportional to the minimum creep rate. **Creep ductility** $\varepsilon_r$ = strain at rupture (5–40 %).

Ex 1. (n) $\dot{\varepsilon} = 10^{-8}\ h^{-1}$ at 100 MPa and $10^{-6}$ at 200 MPa (same $T$) : $n$ = log(100)/log2 = 6.6. At 120 MPa : $\dot{\varepsilon} = 10^{-7}(1.2)^{6.6} = 3.4\times10^{-7}\ h^{-1}$ ⇒ time to 1 % creep = $0.01/3.4\times10^{-7} \approx 3\times10^4$ h.

Ex 2. (Q) $\dot{\varepsilon} = 10^{-9}\ h^{-1}$ at 800 K and $10^{-8}$ at 850 K (same $\sigma$) : $Q = R\,\ln10 /(1/800 - 1/850) = 8.314 \times 2.303 / 7.35\times10^{-5} = 260$ kJ/mol.

## 3. Creep Mechanisms




At temperature $T > 0.4\ T_m$, thermally activated processes (diffusion) allow dislocations and atoms to overcome obstacles. Dominant mechanism depends on stress and temperature.

- **(i) Dislocation glide** (low T/high σ) : dislocations glide on slip planes, blocked by obstacles – exhaustion of mobile dislocations, hardening – responsible for logarithmic / primary creep at low T (< 0.3 $T_m$).
- **(ii) Dislocation climb (power-law, dislocation creep)** : edge dislocations stopped by obstacles (precipitates, other dislocations) climb by absorbing/emitting vacancies and then glide on the new plane – rate controlled by **lattice diffusion** ; gives n = 3–8 :

$$\dot{\varepsilon} = A\ (D\ G\ b\ /\ kT)\ (\sigma/G)^n\ ;\ D = D_0\ exp(-Q/RT)$$

Substructure : subgrains with low-angle boundaries (formed by polygonisation), size ≈ 10 G b/σ.

- **(iii) Diffusional creep** (high T, low σ, fine grains) : stress-directed vacancy flow from boundaries under compression to boundaries under tension, so that grains elongate along the stress axis – n = 1 (Newtonian-viscous) :

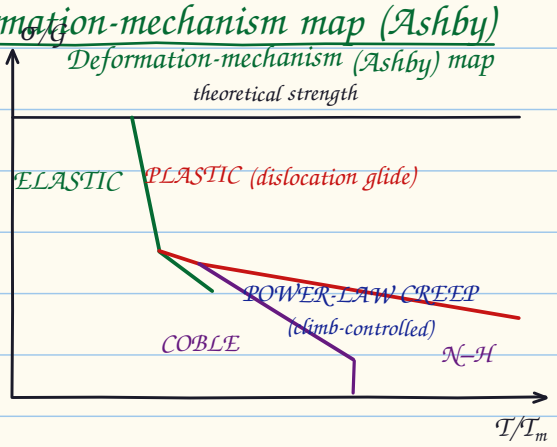
**Nabarro–Herring** (through lattice) : $\dot{\varepsilon} = A_1\ \Omega\ \sigma\ D_l/(k\ T\ d^2)$ **Coble** (along grain boundaries) : $\dot{\varepsilon} = A_2\ \Omega\ \sigma\ \delta\ D_{gb}/(k\ T\ d^3)$

d = grain size ; Ω = atomic volume ; δ = boundary width. Coble dominates at lower temperatures and finer grains. Coarse grains/ single crystals – diffusional creep decreases ⇒ the reason for directional solidification.

- **(iv) Grain-boundary sliding** : grains slide past each other by viscous shear along boundaries (above ≈ 0.5 $T_m$, low σ) – must be accommodated by diffusion or slip at triple points; contributes up to 50 % of total strain ; opens wedge cracks and

cavities (Ashby–Verrall model). **Harper–Dorn** creep ($n \approx 1$) at very low stress and $T > 0.9\ T_m$.

## Deformation-mechanism map (Ashby)

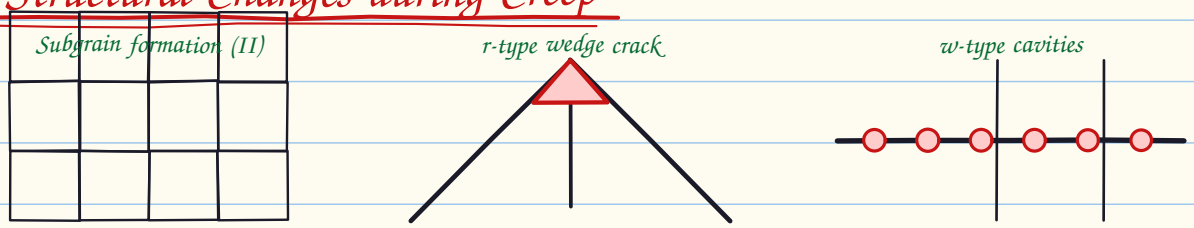



Fields = regions where one mechanism dominates the strain rate for given σ, T, d

Creep : $T > 0.4\ T_m$
Coble (low T, fine d)
N–H (high T, low σ)

- A diagram of normalised stress $\sigma/G$ versus homologous temperature $T/T_m$ divided into fields where a single mechanism dominates the strain rate (elastic, plastic glide, power-law creep, Nabarro–Herring, Coble) with contours of constant $\dot{\varepsilon}$ : lets one read the dominant mechanism and strain rate for given σ, T and grain size d ; used in design of turbine alloys and in geophysics.

## 4. Structural Changes during Creep




size ∝ G b/σ

triple point, high σ, low T

on ⊥ boundaries, low σ, high T
→ link up → intergranular rupture

- **Primary** : rapid dislocation multiplication, tangles, slip bands ; hardening ⇒ $\dot{\varepsilon}$ falls.
- **Secondary** : recovery by climb and cross-slip ⇒ formation of subgrains (cell structure) with low-angle boundaries – dynamic equilibrium ; subgrain size $\lambda \approx 10\ G\ b/\sigma$ (smaller at high σ) ; slip and grain-boundary migration produce **creep-induced grain-boundary serrations** ; slip bands become coarse and straight at higher T.
- **Tertiary** : necking in ductile metals ; internal cavitation and grain-boundary cracking : **r-type (wedge) cracks** at triple points (high stress, lower temperature, caused by GB sliding) ; **w-type (round) cavities** on boundaries perpendicular to the

stress (low stress, high temperature, vacancy condensation) ; cavities grow by diffusion and link up into intergranular cracks.

- **Microstructural degradation** : coarsening (Ostwald ripening) of precipitates (γ′, carbides), rafting of γ′ in superalloys, dissolution or over-ageing, spheroidisation of pearlite, graphitisation, precipitation of embrittling σ-phase, carburisation/oxidation, recrystallisation of cold-worked metal.
- **Creep fracture** : high stress/low T/short time ⇒ transgranular, ductile (cup-and-cone, 20–40 % RA) ; low stress/high T/long time ⇒ intergranular with little ductility (3–10 %), grain-boundary cavities and wedge cracks. Fracture-mechanism maps (Ashby) show these fields.

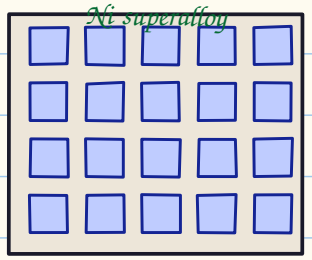
## 5. Metallurgical Factors Affecting Creep

| Factor | Effect | Practice |
|---|---|---|
| High melting point | lower diffusivity, high modulus ⇒ better ; creep resistance ∝ $T/T_m$ | Ni, Co, Nb, Mo, W, Ta |
| Crystal structure | close-packed FCC (low diffusivity) better than BCC (open, fast diffusion) ; low stacking-fault energy hinders climb | austenitic steels, Ni alloys |
| Grain size | coarse grain better at high T (less GB sliding/diffusion) ; fine grain better below the equicohesive temperature $T_E$ | DS and SX blades |
| Solid-solution | large atoms (Mo, W, Re) slow diffusion, lower SFE | Ni-Mo-W-Re |
| Precipitates (γ′, carbides) | coherent ordered particles block climb ; $M_{23}C_6$, $M_6C$ carbides at boundaries stop GB sliding | Ni superalloys, Cr–Mo–V steels |
| Dispersions ($ThO_2$, $Y_2O_3$) | stable to ≈ 0.9 $T_m$ – no coarsening | ODS alloys |
| Alloying with B, Zr, Hf | segregate to boundaries : ↑ creep ductility, stops cavity growth | superalloy boundaries |

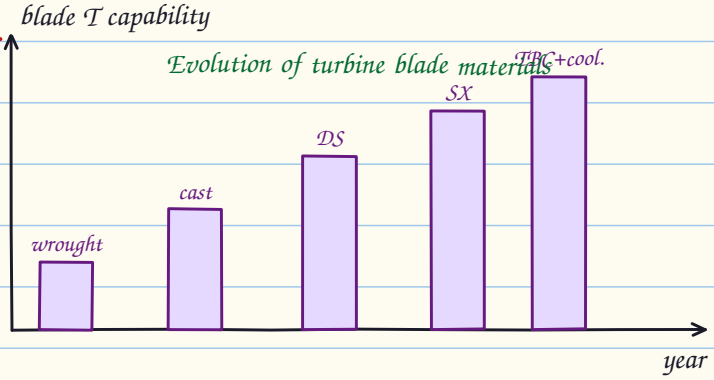
| Factor | Effect | Practice |
|---|---|---|
| Impurities (S, P, Sn, Pb) | segregate to boundaries ⇒ embrittle | clean melts, vacuum casting |
| Prior cold work | may raise creep resistance at low $T$ but recovers at high $T$ | not relied on |
| Heat treatment | solution + double ageing – optimum precipitates | $\gamma'$ size & distribution |
| Environment | oxidation, hot corrosion, creep-oxidation cracking | Cr, Al coatings |

- **Equicohesive temperature** $T_E$ : temperature where grain boundaries and grain interiors have equal strength ; below $T_E$ ($\approx 0.5\ T_m$) fine grains stronger (Hall–Petch), above it boundaries are the weak path (coarse grains / aligned boundaries better).

## 6. High-Temperature Alloys




$\gamma'$ [$Ni_3(Al,Ti)$] cuboids in $\gamma$ matrix
60–70 % $\gamma'$, size ≈ 0.3–0.5 µm



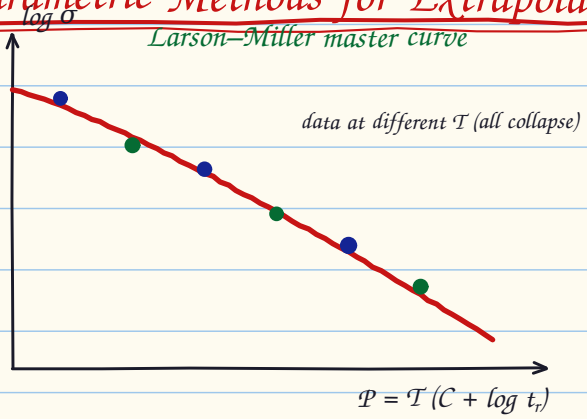

Requirements : high-temperature strength and creep resistance, oxidation / hot-corrosion resistance, microstructural stability, fatigue and thermal-shock resistance, workability / castability, cost.

| Class | Examples | Service limit |
|---|---|---|
| Low-alloy ferritic steels | 1–1.25Cr–0.5Mo, 2.25Cr–1Mo (boilers, pipes) | ≈ 565 °C |
| Advanced 9–12Cr steels | 9Cr–1Mo–V (P91), P92 | ≈ 600–620 °C |
| Austenitic stainless | 304H, 316, 321, 347, 310 | ≈ 650–700 °C |
| Iron-base superalloys | A-286, Incoloy 800 | ≈ 700 °C |

| Class | Examples | Service limit |
|---|---|---|
| Nickel-base superalloys | Inconel 718, 625, Waspaloy, Rene 41/80, IN-738, MAR-M247, CMSX-4 | up to 1100 °C (0.85 $T_m$) |
| Cobalt-base | Haynes 25 (L-605), Stellite, X-40 | vanes, hot corrosion ≈ 1000 °C |
| Refractory metals | Nb, Mo, Ta, W alloys (C-103, TZM) | > 1200 °C (oxidation protection needed) |
| Intermetallics / ceramics | TiAl, $Ni_3Al$, $MoSi_2$ ; SiC, $Si_3N_4$, CMC (SiC/SiC) | up to 1300–1500 °C |

- **Ni-base superalloy microstructure** : FCC γ matrix (Ni–Cr–Co solid solution with Mo, W, Re) + coherent ordered γ′ $Ni_3(Al,Ti)$ cuboids (60–70 vol %) that strengthen up to 0.8 $T_m$ (strength rises with T – anomalous yield) ; carbides (MC, $M_{23}C_6$) on grain boundaries ; B, Zr add boundary cohesion ; TCP phases (σ, μ) avoided.
- **Processing evolution** : wrought → conventional cast → **directionally solidified (DS)** columnar grains parallel to stress (no transverse boundaries) → **single crystal (SX)** (no boundaries ⇒ no B, C, Zr needed – CMSX-4 with Re) ; each step ≈ +20–30 °C ; **ODS** alloys (MA6000) ; powder-metallurgy discs (René 95) ; film-cooled blades, thermal barrier coatings (YSZ over MCrAlY bond-coat) add 100–150 °C ; **aluminide / MCrAlY coatings** for oxidation.
- Selection : steam pipes (P91), gas turbine blades (SX Ni-base + TBC), discs (IN-718/ René 95), combustors (Hastelloy X, Haynes 230), exhaust/automotive turbo (Inconel 713), heating elements (Ni–Cr 80/20 Nichrome, Kanthal Fe–Cr–Al), rocket nozzles (Nb, C/C).

## 7. Parametric Methods for Extrapolating Creep Data




$P = T\,(C + \log t_r)$

$T$ in K, $t_r$ in hours, $C \approx 20$

Sherby–Dorn : $P_{SD} = \log t_r - 0.434Q/RT$

Manson–Haferd : $(\log t_r - \log t_a)/(T - T_a)$

Monkman–Grant : $\dot{\varepsilon}_s \cdot t_r = C_{MG}$

Long-life data ($10^5$ h) cannot be obtained by direct testing ⇒ short-time high-temperature tests are extrapolated using **time–temperature parameters** :

**Larson–Miller : $P = T\,(C + \log t_r)$ ($T$ in K, $t_r$ in h, $C \approx 20$)**

- Principle : $\log t_r + \text{const} = Q/(2.3RT)$ – same rupture life at different ($T$, $t$) combinations at the same stress ⇒ a single **master curve** $\log \sigma$ vs $P$ ; read $P$ at the design stress and compute the life at service $T$. Others : Sherby–Dorn, Manson–Haferd, Orr–Sherby–Dorn.

Ex 3. A stress gives $t_r$ = 1000 h at 900 K : $P = 900(20 + 3) = 20\,700$. At 850 K : $\log t_r = 20\,700/850 - 20 = 4.35 \Rightarrow t_r = 2.25\times10^4$ h. To last 25 000 h ($\log$ = 4.40) : $T = 20\,700/24.4 = 848$ K = 575 °C.

Ex 4. Steel : at $P$ = 18 000 ($T$ in K) allowable stress read from the master curve ; for $T$ = 700 K life $\log t_r = 18\,000/700 - 20 = 5.71 \Rightarrow 5.1\times10^5$ h.

## 8. Stress-Rupture Testing

- **Stress-rupture (creep-rupture) test** : specimens loaded at constant load/temperature until fracture ; records time to rupture $t_r$, elongation and reduction of area at fracture, sometimes minimum creep rate. Higher stresses than creep tests, shorter times (up to $10^3$–$10^5$ h).
- Result presented as **$\log \sigma$ vs $\log t_r$** at several temperatures – straight lines (parallel) ; a break in slope at long times indicates a change in mechanism : over-ageing, recrystallisation, oxidation, transition from transgranular to intergranular failure ; notched specimens test **notch sensitivity** (notch strengthening vs weakening). The **rupture strength** (e.g. stress for rupture in 1000

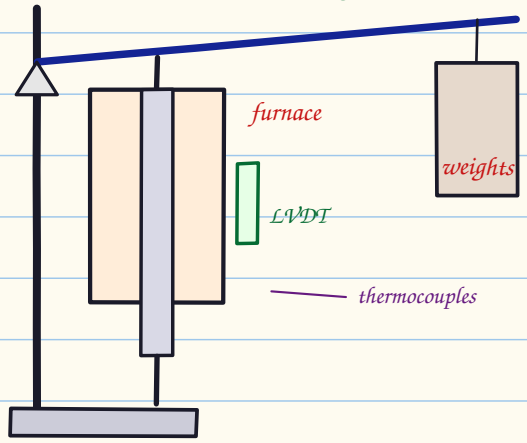
/ 10 000 / 100 000 h at T) is the design property ; also stress for 1 % creep in $10^4$ h.

- **Procedure** (ASTM E139, E292, ISO 204) : round specimen (gauge dia 6.35 mm, gauge length 25–50 mm, sometimes notched), thermocouples attached, furnace set ±2–3 °C (to 1000 °C ; ±5 up to 1200), allowed to equilibrate, load applied smoothly via lever/weights, elongation measured, timer stops at rupture (microswitch), fractography and metallography afterwards.
- **Design stresses** : allowable stress = lowest of ($\sigma_y$/1.5, 100 % average stress for $10^5$ h rupture ÷ 1.5 (or 0.67 of mean), stress for 1 % creep in $10^5$ h). (ASME code.)

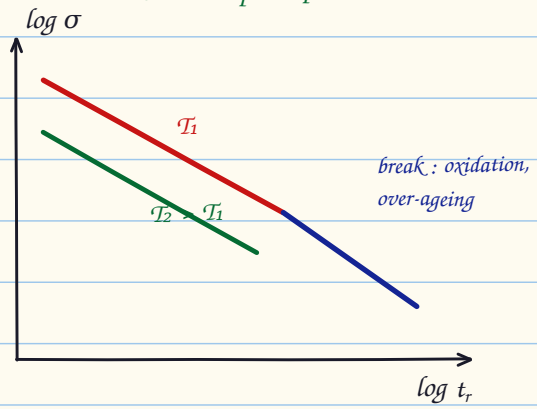
Ex 5. A steel steam pipe designed for $10^5$ h at 550 °C (P from master curve) rupture strength 90 MPa; allowable design stress = 0.67 × 90 = 60 MPa (or the stress for 1 % creep in $10^5$ h, whichever is lower).

## 9. Creep Testing Machines

Lever-arm creep testing machine




Stress–rupture plot




### Components of a creep testing machine

- **Loading system** : dead-weight lever-arm (lever ratio 3 : 1 to 20 : 1 ; load pan with weights) ; spring/hydraulic or screw systems (for stress-relaxation and constant-rate tests) ; servo-hydraulic frames ; for **constant true stress** a cam, lever-contour (Andrade–Chalmers) or electronic controller adjusts the load as the area changes.
- **Furnace** : split tubular resistance furnace (Kanthal / Pt / Mo wire) ; multiple zones ; PID temperature controller ; temperature uniform within ±2 °C over the gauge length at ≤ 1000 °C ; thermocouples (type K/N/S) on the specimen

shoulders ; inert or vacuum atmosphere for refractory metals.

- **Extensometry** : specimen shoulders carry arms outside the furnace ⇒ *LVDT / capacitive / optical* extensometer, strain sensitivity $10^{-6}$–$10^{-5}$ ; data logger plotting ε–t. Alignment grips / universal joints to avoid bending ; pull-rods of heat-resistant alloy.
- **Test types** : **creep test** (ε–t, long term 1000–$10^5$ h, small stress to get minimum creep rate and 0.1–1 % strains), **stress-rupture test, stress-relaxation test** (strain fixed, stress decays – bolts), **step-load / step-temperature tests** for n and Q, **constant strain-rate, creep-fatigue,** small-punch and impression creep for in-service components, miniature specimens.
- **Precautions** : constant temperature, accurate axial alignment, no shock loading, correction for thermal expansion and drift ; multiple specimens in line (multi-station racks) ; tests run 24 h/day. Accelerated testing : raise stress and/or T and use parameters (LMP).
- **Short-time tension tests at high temperature** – determine hot yield and tensile strength, ductility ; hot hardness test.

## 10. Additional Solved Problems

P1. The steady creep rate of a metal is $2\times10^{-7}\ h^{-1}$ at 100 MPa and 600 °C ; n = 5. Find the rate at 150 MPa : $(1.5)^5 = 7.59$ ⇒ $1.5\times10^{-6}\ h^{-1}$.

P2. Q = 250 kJ/mol ; rate at 800 K is $10^{-8}\ h^{-1}$. At 850 K : ratio = exp[(250 000/8.314)(1/800 − 1/850)] = exp(30 070 × $7.35\times10^{-5}$) = exp(2.21) = 9.1 ⇒ $9.1\times10^{-8}\ h^{-1}$.

P3. A bar length 500 mm creeps at $5\times10^{-8}\ s^{-1}$ for 1000 h : strain = $5\times10^{-8}\times3.6\times10^{6}$ = 0.18 (18 %) – too much ; a rate of $10^{-10}\ s^{-1}$ ⇒ $3.6\times10^{-4}$ (0.036 %) in 1000 h.

P4. Monkman–Grant : $\dot{\varepsilon}_s = 10^{-6}\ h^{-1}$, $C_{MG} = 0.05$ ⇒ $t_r = 0.05/10^{-6} = 5\times10^{4}$ h.

P5. LMP (C = 20) : stress rupture at 700 °C (973 K) for 1000 h : P = 973 × 23 = 22 380. Life at 650 °C (923 K) = $10^{22\,380/923\ -\ 20} = 10^{4.25} = 1.8\times10^{4}$ h.

## Short answers – Quick Revision

- *Why does creep occur only at high temperature?* Needs thermally activated diffusion (climb, vacancy flow, GB sliding) – significant above 0.4 $T_m$.

- *Why is stage II called steady state?* Work hardening equals recovery ⇒ constant dislocation structure and rate.
- *Why are single crystals used for turbine blades?* No grain boundaries ⇒ no GB sliding, cavitation or diffusional creep.
- *Difference between creep and stress relaxation?* Creep – strain increases at constant stress ; relaxation – stress decays at constant strain.
- *What does the LMP do?* Relates rupture life, stress and temperature in one master curve for extrapolation.

## 11. Quick Formula Sheet – Unit V

| Item | Formula |
|---|---|
| Homologous temperature | $T/T_m > 0.4 \Rightarrow$ creep |
| Steady creep rate | $\dot{\varepsilon}_s = A\sigma^n \exp(-Q/RT)$ |
| Garofalo | $\varepsilon = \varepsilon_0 + \varepsilon_t(1 - e^{-rt}) + \dot{\varepsilon}_s t$ |
| Nabarro–Herring | $\dot{\varepsilon} \propto \sigma D_l / d^2$ |
| Coble | $\dot{\varepsilon} \propto \sigma D_{gb} / d^3$ |
| Monkman–Grant | $\dot{\varepsilon}_s t_r = C$ |
| Larson–Miller | $P = T(C + \log t_r)$ , $C \approx 20$ |
| Activation energy | $Q = R\,\ln(\dot{\varepsilon}_2/\dot{\varepsilon}_1)/(1/T_1 - 1/T_2)$ |

## 12. Important Questions

- Define creep. Draw a typical creep curve and explain the three stages with the underlying mechanisms.
- Discuss the effect of stress and temperature on the creep curve ; Norton law and activation energy for creep.
- Explain the creep mechanisms : dislocation glide/climb, diffusional (Nabarro–Herring, Coble), grain-boundary sliding ; deformation-mechanism maps.
- Describe the structural changes during creep ; creep fracture (wedge cracks, cavities).
- Discuss the metallurgical factors affecting creep resistance ; why single crystals / DS blades?

- Write notes on high-temperature alloys (steels, Ni-base superalloys) and their strengthening.
- Explain the stress-rupture test and Larson–Miller parameter. Describe a creep testing machine with a neat sketch.